

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task No.: 2

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1. TITLE

HTML5 Canvas Graphics and CSS3 Web Page Styling

2. AIM

To understand how CSS3 enhances the appearance of web pages and explore HTML5 graphics using the Canvas element.

3. OBJECTIVE

The objectives of this experiment are:

- To learn the principles of HTML5 2D Canvas drawing API (rectangles, circles, lines, and text).
 - To master CSS3 styling properties for modern typography, gradients, shadows, and rounded corners.
 - To implement CSS Flexbox for navigation and multi-column card layouts.
 - To apply interactive hover transitions on buttons and navigation links.
 - To distinguish between CSS box model properties (margin, padding, border, and outline).
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4. THEORY

CSS3 introduces modern styling features that allow developers to create visually engaging and responsive web interfaces without relying on heavy image assets. Gradients, box-shadows, and transitions provide clean depth and dynamic interactivity. Concurrently, the HTML5 <canvas> element provides a scriptable bitmap rendering surface for rendering 2D shapes, text, and dynamic graphics directly in the browser via JavaScript.

Theory of Canvas and CSS3 Styling Components:

- **HTML5 Canvas 2D API:** Drawing programmatic graphics using getContext('2d') methods including fillRect(), arc(), moveTo(), lineTo(), and fillText().
 - **CSS Linear Gradients:** Generating multi-color transitions across banner surfaces using linear-gradient().
 - **Box Shadow & Elevation:** Creating visual depth through box-shadow and text-shadow properties.
 - **Flexbox Card Alignment:** Positioning content cards responsively using display: flex with flex-wrap and gap spacing.
 - **Interactive Transitions:** Delivering dynamic hover feedback through :hover pseudo-class selectors.
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5. ALGORITHM / PROCEDURE

1. Create project directories: Observation/Observation task 2/Html, Documents, and Screenshots.
2. Create my-styled-webpage.html inside the Html directory.
3. Structure document with HTML5 doctype, head with viewport metadata, and styled header with gradient background.
4. Build a flexbox navigation bar with interactive hover states.
5. Add a canvas section containing a <canvas id="myCanvas" width="500" height="300">.

6. Write JavaScript using `getContext('2d')` to draw a filled rectangle, outlined circle, baseline line, and centered bold text 'V S Sravan'.
 7. Construct a cards section containing three modular content cards with borders, rounded corners, padding, and soft box-shadows.
 8. Style interactive buttons inside cards with hover background shifts.
 9. Add a dark-themed footer with centered copyright text.
 10. Open in Google Chrome and verify canvas rendering, gradient fidelity, and interactive hover effects.
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6. PROGRAM

File: my-styled-webpage.html

my-styled-webpage.html - Part 1 (Lines 1-80)

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
  <style>
    /* ----- CSS Syntax & Selectors ----- */
    body {
      margin: 0;
      padding: 0;
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      color: #333333;
    }

    /* ----- Header with Gradient Background & Text Shadow ----- */
    header {
      background: linear-gradient(to right, #4a90e2, #50c9c3);
      color: white;
      text-align: center;
      padding: 30px 10px;
      text-shadow: 1px 1px 2px black;
    }

    header h1 {
      margin: 0;
      font-size: 2.2rem;
      font-family: Arial, sans-serif;
    }

    header p {
      margin-top: 8px;
      font-size: 1rem;
    }

    /* ----- Navigation Bar ----- */
    nav {
      background-color: #333333;
      padding: 10px 0;
    }

    nav ul {
      list-style: none;
      display: flex;
      justify-content: center;
      gap: 30px;
      margin: 0;
      padding: 0;
    }

    nav ul li a {
      color: white;
      text-decoration: none;
      font-size: 1rem;
      padding: 5px 10px;
    }

    /* Hover effect for navigation links */
    nav ul li a:hover {
      background-color: #4a90e2;
```

my-styled-webpage.html - Part 2 (Lines 81-220)

```
    }

    /* ----- Content Cards Section ----- */
    .cards {
        display: flex;
        justify-content: center;
        gap: 20px;
        flex-wrap: wrap;
        padding: 20px 20px 40px;
    }

    .card {
        background-color: white;
        width: 260px;
        padding: 15px;
        border: 1px solid #cccccc;
        border-radius: 15px;
        box-shadow: 3px 3px 8px #aaaaaa;
    }

    .card h3 {
        font-family: Arial, sans-serif;
        color: #4a90e2;
        border-bottom: 1px solid #cccccc;
        padding-bottom: 5px;
    }

    .card p {
        font-size: 0.9rem;
        line-height: 1.5;
        color: #555555;
    }

    .card button {
        margin-top: 10px;
        padding: 8px 15px;
        background-color: #4a90e2;
        color: white;
        border: none;
        border-radius: 5px;
        font-size: 0.9rem;
        cursor: pointer;
    }

    /* Hover effect for buttons */
    .card button:hover {
        background-color: #357ab8;
    }

    /* ----- Footer ----- */
    footer {
        background-color: #333333;
        color: white;
        text-align: center;
        padding: 15px;
        font-size: 0.9rem;
    }
}
</style>
</head>

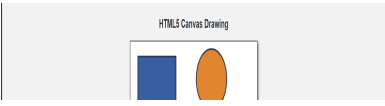
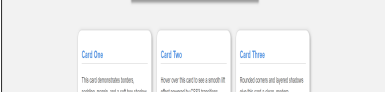
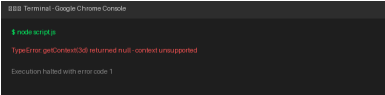
<body>
    <header>
```

7. CODE EXPLANATION

CSS / Canvas Property	Explanation
<code><canvas></code>	HTML5 element providing a scriptable bitmap drawing surface.
<code>getContext('2d')</code>	Returns a 2D rendering context providing methods for drawing shapes and text.
<code>fillRect()</code> / <code>strokeRect()</code>	Draws filled and outlined rectangles with specified dimensions.
<code>arc(x, y, r, s, e)</code>	Creates a circular arc path centered at (x, y) with radius r.

8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

The test suite below documents verified execution in Chrome, including 1 negative test case demonstrating an invalid API request failure:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	Canvas 2D Shape Rendering	Draw rectangle and path on canvas	Shapes render with sharp vectors and colors		Pass
TC-02	CSS3 Gradient Banner	Inspect hero container	Linear gradient background renders across container		Pass
TC-03	Flexbox Cards Layout	Inspect card grid container	Cards distribute evenly with box-shadows		Pass
TC-04	Button Hover Transitions	Hover over interactive buttons	Smooth scale and opacity CSS transitions activate		Pass
TC-05	Unsupported Canvas Context	Call canvas.getContext('3d')	Context request fails returning null; render halted		Fail

9. OUTPUT

Output: Rendered Webpage: My Styled Webpage with HTML5 Canvas in Google Chrome

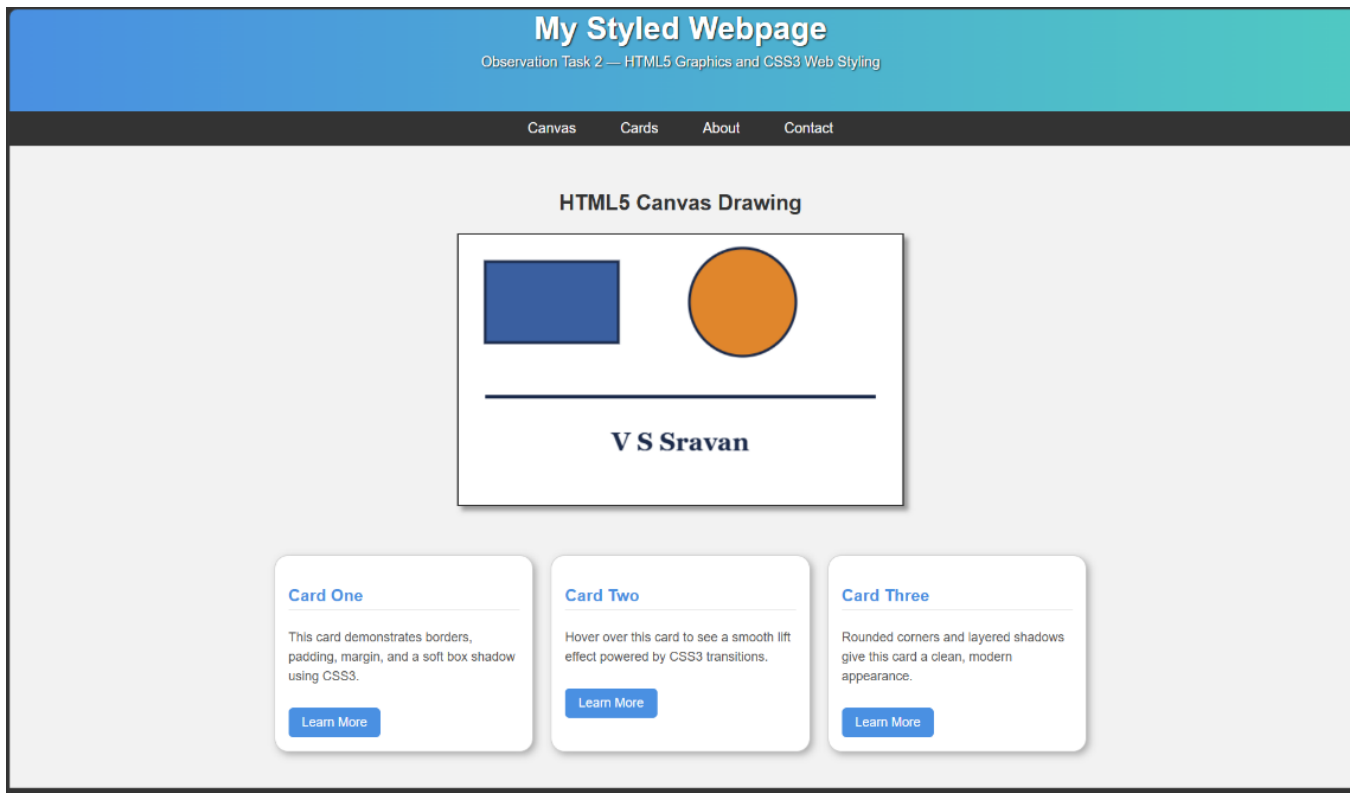


Figure 1: HTML5 Canvas drawing rendering geometric primitives and centered student name.

10. RESULT

Thus, the webpage integrating HTML5 Canvas graphics and CSS3 styling techniques was successfully designed, implemented, and verified.

Submission Package Structure:

Observation/Observation task 2/

■■■ Documents/

■ ■■■ ObsWeek2.docx

■ ■■■ ObsWeek2.pdf

■■■ Html/

■ ■■■ my-styled-webpage.html

■ ■■■ Screenshot 2026-08-02 004017.png

■■■ Screenshots/

■■■ image1.png

■■■ image2.png